

MUHAMMAD HAIDER ZAIDI

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🌐 github.com/chefhaider 🏠 Bavaria, Germany 🎂 21.06.2000

Technical Skills

Programming: Python, SQL, Javascript

AI/ML: GenAI, AgenticAI, LLM-APIs, Prompt Engineering, Local Models, Embeddings, NLP, Machine Learning, Deep Learning (Tensorflow, Keras, PyTorch), AI/ML Architectures (CNN, YOLO, Transformers)

MLOps/Deployment: Docker, Kubernetes, AWS SageMaker, FastAPI, CI/CD, Performance Monitoring, Model Tuning, Real-time Systems

Data Engineering/Databases: ETL/ELT Pipelines, Feature Engineering, Data Processing, Relational Databases (SQL), Big Data Analytics

Experience

German National Metrology Institute (PTB)

March 2024 – Present

Student Assistant Researcher

Berlin, DE

- Designed, evaluated, and optimized deep learning models for metrological applications, enhancing model performance and training efficiency.
- Explored novel KI architectures and physics-informed hybrid loss functions, demonstrating commitment to innovation and improving model accuracy and stability.
- Collaborated effectively within interdisciplinary teams for a publication at FLOMEKO conference, showcasing technical consultation and presentation skills.
- Developed dashboards for data visualization and continuous model performance monitoring and evaluation, contributing to optimization efforts.

Folio3

March 2022 – Nov 2023

Machine Learning Engineer

California, US (remote)

- Improved vitiligo segmentation accuracy to 94% and increased inference speed by 5x using advanced AI algorithms, deploying scalable KI-solutions with Docker and FastAPI.
- Developed a real-time tennis foul detection system utilizing TensorRT, optimizing ML-model performance and reducing compute costs by 6x.
- Managed comprehensive MLOps pipelines on AWS SageMaker, proactively addressing YOLO model data drift and maintaining Kubernetes infrastructure, ensuring robust CI/CD-pipelines and operational quality.
- Conducted educational sessions on machine learning and big data, sharing expertise and fostering knowledge transfer within interdisciplinary teams and external stakeholders.
- Developed Proof-of-Concepts (POCs) using YOLO, few-shot learning, and CNN architectures for client demonstrations, evaluating new technologies and identifying AI use-cases.

Manda GmbH

March 2024 - June 2024

Werkstudent Python Developer

Nürnberg, DE

- Engineered full-stack solutions for web-based AI services using Flask, HTML, and CSS, translating KI-technologies into practical solutions.
- Developed and implemented an AI chatbot application leveraging OpenAI API and relevant LLM tools/frameworks, showcasing direct access to state-of-the-art LLMs and Prompt Engineering.

Freelance

July 2020 – March 2022

Data Engineer

Remote

- Engineered robust web scrapers and ETL/ELT pipelines for efficient extraction and processing of structured and unstructured large datasets.
- Automated data ingestion and cleaning workflows, performing data analysis and preparing large data quantities for client analytics use-cases.

Projects

CAFA 6: Protein Function Prediction

2022

Deep Learning Competition

- Architected a weighted-probability ensemble integrating DPFunc and DeepGOPlus, leveraging AlphaFold for feature extraction and mitigating taxonomic bias in complex datasets.
- Increased model accuracy by 4% through custom feature engineering and data analysis across MF, CC, and BP namespaces.

Razanlytics: Real-time Bitcoin Price Forecasting

2022

Thesis Project

- Developed and implemented a real-time Bitcoin price forecasting system using advanced time-series Machine Learning models.
- Integrated external APIs for data ingestion and performed continuous model evaluation and tuning.

Saaz: AI Self-Playing Chitralli Sitar

2023

Signal Processing, Music Generation

- Designed real-time signal processing system integrating temporal dependencies for innovative music generation KI-solutions.
- Implemented transformer-based models and NLP techniques for sequence prediction in audio generation tasks, demonstrating expertise in cutting-edge GenAI applications.

Hugging Face

Open-Source

Contributor

- Co-authored Zero-shot Image Classification technical documentation and implementation page on the Hugging Face Hub, contributing to open-source AI knowledge-bases and demonstrating understanding of off-the-shelf solutions.

Education

Friedrich Alexander Universität

Present

Master of Science in Artificial Intelligence

Erlangen, DE

- Specialized in deep learning for computer vision and spatio-temporal modeling
- Relevant coursework: Advanced Neural Networks, Computer Vision, Autonomous Systems

National University of Computer and Emerging Sciences Aug 2018 – June 2022

Bachelor of Science in Computer Science

Karachi, PK

- Dean's List of Honor
- Specialized in Machine Learning and Computer Vision with focus on time-series analysis

Certifications

PyTorch: Deep Neural Networks with PyTorch | **Coursera:** TensorFlow Developer Certificate
| **Databricks:** Lakehouse Fundamentals | **Datacamp:** Computer Vision in Python